

1. Alert

An alert is like an automatic alarm that a security system sends to the security team when it notices something unusual or dangerous happening on the network. Just like how a smoke detector makes noise when it detects smoke, a SIEM alert notifies the security team the moment it detects suspicious activity such as too many failed logins, unauthorized access, or unusual data transfer. It is basically the system saying "hey something is wrong, come and check this."

2. Incident Response Steps

Step 1: Preparation Before any attack happens the security team prepares by setting up tools, writing security policies, and training staff on what to do during an attack. This step is done in advance so the team is ready when something goes wrong.

Step 2: Identification When an alert is triggered the team first checks whether it is a real threat or a false alarm. They look at the logs, analyze the alert details, and confirm whether an actual security incident has taken place.

Step 3: Containment Once the threat is confirmed the team immediately tries to stop it from spreading. For example if a system is infected they disconnect it from the network so the attack does not reach other systems.

Step 4: Eradication After containing the threat the team finds the root cause and completely removes it from the system. This could mean deleting malware, removing a backdoor, or closing a vulnerability that the attacker used to get in.

Step 5: Recovery After removing the threat the affected systems are restored and brought back online carefully. The team monitors everything closely to make sure the attack does not happen again.

Step 6: Lessons Learned After everything is resolved the team holds a meeting to discuss what happened, how it was handled, and what can be done better next time. This step helps improve the security of the organization for the future.

3. Real World Example

Scenario: Ransomware Attack on a Hospital

In 2020 a hospital in Germany received a ransomware attack where the attackers encrypted all the hospital's data and demanded payment to unlock it. Here is how the incident response steps were applied in this real situation:

During identification the IT team noticed that files across the network were suddenly getting encrypted and systems were becoming unavailable. The SIEM tool generated critical alerts about unusual file modification activity happening on hundreds of systems simultaneously.

During containment the team immediately shut down and disconnected all affected systems from the network to prevent the ransomware from spreading to other departments and medical devices.

During eradication the cybersecurity team identified the ransomware strain, found the entry point which was a phishing email opened by an employee, and completely wiped and rebuilt the infected systems.

During recovery the hospital restored all its data from clean backups that were not affected by the attack and slowly brought systems back online one by one after confirming they were safe.

During lessons learned the hospital introduced mandatory cybersecurity awareness training for all staff, improved their email filtering system to block phishing emails, and set up better backup systems to prevent future data loss.

This real world example shows how important it is to have a proper incident response plan because without it the hospital could have permanently lost all patient data and faced life threatening consequences.